

Archaeologist

16+3 · WEIGHT 1 · LONGEST (Kanta Saito)

CLEAR priors → inventum (image JEPA + multi-block) → V-JEPA → V-JEPA
2-AC robotics. Label every lineage claim CLEAR vs TENTATIVE.

Spine: Predict embeddings, not pixels · multi-block + EMA + narrow predictor · JEA|MAE|I-JEPA ·
→V-JEPA→2-AC (cite 2-AC for robots) · 54.2 / 66.9>>40.7 / ~10× / 72.4

Assran et al. · Meta FAIR · Predict embeddings, not pixels · facebookresearch/ijepa

Talk arc — densest lineage pack

Priors → inventum → subsequent → triangle → robotics deepen

01-03

Spine + CLEAR priors

AMI JEPA · EBM · Siamese · MAE · CPC

04-07

Inventum + paper figs

multi-block · EMA · narrow predictor · Fig 2/3/4

08-10

Subsequent + foils

V-JEPA → 2-AC CLEAR · DINOv2/ π_0 TENTATIVE

11-14

Triangle + world-model

consistency · RCDM · cite 2-AC for robot SR

15-17

Must numbers + close

tattoo · CLEAR/TENTATIVE exam bite

Shared spine — tattoo every deck

Six claims that must land in every seat

1 Predict embeddings, not pixels.

2 Multi-block masking + EMA target + narrow predictor.

3 Taxonomy: contrastive JEA | MAE pixels | I-JEPA emb.

4 CLEAR lineage → V-JEPA → V-JEPA 2-AC (cite 2-AC for robot claims).

5 Must: multi-block 54.2 · emb>>pixels 66.9 vs 40.7 · ~10× vs MAE · Clevr Dist 72.4.

6 Headlines: 79.3% · 73.3% 1% · 81.1%/77.3% @448 · 16×A100 <72h / <1200 GPU-h.

CLEAR priors — cited / direct parents

Every row is in-paper citation or direct method family

LeCun AMI 2022 [48]	Names JEPA; predictive embeddings + AMI agenda	CLEAR
LeCun EBM tutorial [49]	Energy assign compatible/incompatible — Fig. 2 framing	CLEAR
Predictive coding [59,31]	Rao & Ballard / Friston — predict-missing-input motivation	CLEAR
Siamese SSL	BYOL / DINO / iBOT / SimCLR / MoCo — EMA / stop-grad collapse tricks	CLEAR
MAE / BEiT / SimMIM	Generative MIM foil; sparse-encoder trick reused	CLEAR
data2vec / CAE	Closest: predict representations of masked patches	CLEAR
ViT / DeiT	Backbone	CLEAR
CPC [55,39]	Contrastive local predictive coding — foil (I-JEPA non-contrastive)	CLEAR

This paper — CLEAR inventum

First image instantiation of JEPA

Image JEPA

Predict target-block embeddings from one context block — loss in representation space.

Multi-block mask

M=4 semantic targets (0.15–0.2) + sparse informative context (~ 0.25) — key inductive bias.

EMA + narrow predictor

BYOL-style EMA target; predictor emb=384; positional mask tokens.

No view augs

Single view; no color jitter / multi-crop SimCLR stack — mask replaces invariance prior.

Dual competence

Strong linear probes AND local tasks (Clevr Dist 72.4 $\gg$ DINO 53.4).

Compute win

ViT-H/14 < 1200 GPU-h · 16×A100 < 72h · $\sim 10\times$ vs MAE H/14.

Taxonomy — Fig. 2 (CLEAR)

JEA contrastive | Generative MAE pixels | JEPA embeddings

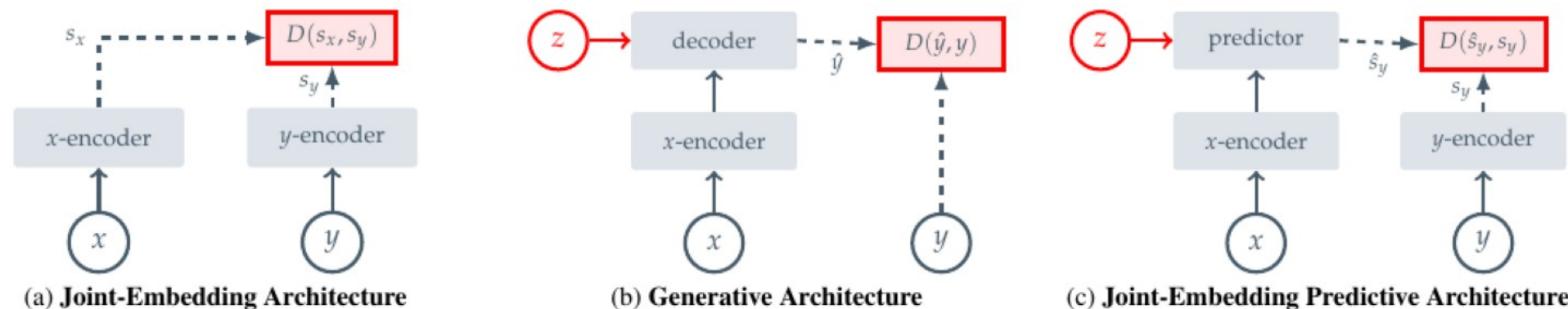


Figure 2. Common architectures for self-supervised learning, in which the system learns to capture the relationships between its inputs. The objective is to assign a high energy (large scaler value) to incompatible inputs, and to assign a low energy (low scaler value) to compatible inputs. **(a)** Joint-Embedding Architectures learn to output similar embeddings for compatible inputs x, y and dissimilar embeddings for incompatible inputs. **(b)** Generative Architectures learn to directly reconstruct a signal y from a compatible signal x , using a decoder network that is conditioned on additional (possibly latent) variables z to facilitate reconstruction. **(c)** Joint-Embedding Predictive Architectures learn to predict the embeddings of a signal y from a compatible signal x , using a predictor network that is conditioned on additional (possibly latent) variables z to facilitate prediction.

resulting representations are typically of a lower semantic level and underperform invariance-based pretraining in off-the-shelf evaluations (e.g., linear-probing) and in transfer

approaches on semantic tasks and achieves better performance on low-level visions tasks such as object counting and depth prediction (Sections 5 and 6). By

Paper Fig. 2 — EBM framing. I-JEPA = (c) predict $\hat{s}_y$ in embedding space, not pixels.

Architecture — Fig. 3 (CLEAR inventum)

Context encoder · EMA target · narrow predictor · L2 in emb space

Generative Architectures. Reconstruction-based methods for self-supervised learning can also be cast in the framework of EBMs using Generative Architectures; see Figure 2b. Generative Architectures learn to directly reconstruct a signal y from a compatible signal x , using a decoder network that is conditioned on an additional (possibly latent) variable z to facilitate reconstruction. In the context of image-based pretraining, one common approach in computer vision is to produce compatible x, y pairs using masking [9, 38] where x is a copy of the image y , but with some of the patches masked. The conditioning variable z then corresponds to a set of (possibly learnable) mask and position tokens, that specifies to the decoder which image patches to reconstruct. Representation collapse is not a concern with these architectures as long as the informational capacity of z is low compared to the signal y .

Joint-Embedding Predictive Architectures. As shown in Figure 2c, Joint-Embedding Predictive Architectures [48]

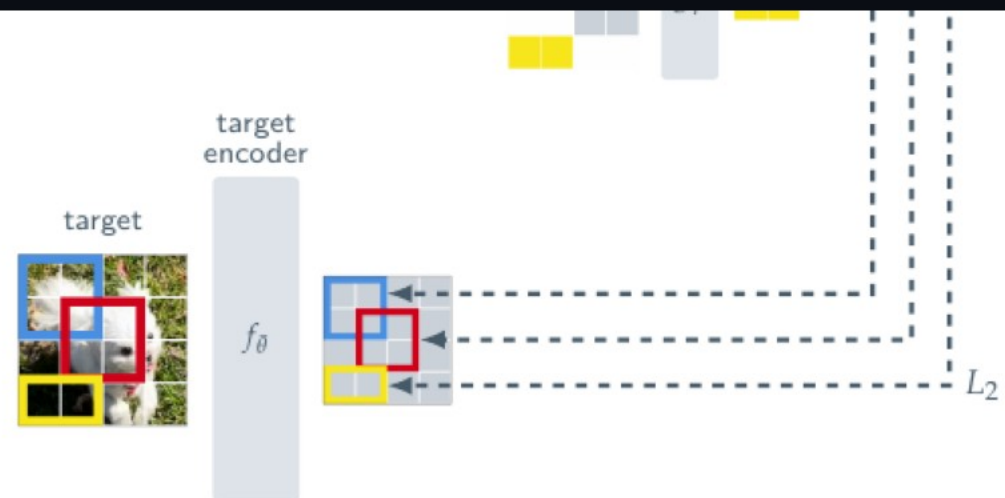


Figure 3. **I-JEPA.** The Image-based Joint-Embedding Predictive Architecture uses a single context block to predict the representations of various target blocks originating from the same image. The context encoder is a Vision Transformer (ViT), which only processes the visible context patches. The predictor is a narrow ViT that takes the context encoder output and, conditioned on positional tokens (shown in color), predicts the representations of a target block at a specific location. The target representations correspond to the outputs of the target-encoder, the weights of which are updated at each iteration via an exponential moving average of the context encoder weights.

Paper Fig. 3 — predictor conditioned on positional mask tokens = latent spatial world-model stub.

Multi-block masking — Fig. 4 (CLEAR)

$4 \times \text{Block}(0.15, 0.2)$ targets · context $\text{Block}(0.85, 1.0) \times \text{complement}$ · T6 $\rightarrow$ 54.2



Figure 4. Examples of our context and target-masking strategy. Given an image, we randomly sample 4 target blocks with scale in the range $(0.15, 0.2)$ and aspect ratio in the range $(0.75, 1.5)$. Next, we randomly sample a context block with scale in the range $(0.85, 1.0)$ and remove any overlapping target blocks. Under this

parameter
positional
for M ta
time con
target-bl
dictions

Loss. T
the predi
get patch

$$\frac{1}{M} \sum_{i=1}^M$$

The para
are learn
parameter
nential m
The use

Paper Fig. 4 — remove context-target overlap $\rightarrow$ sparse yet informative context (avg ratio ~ 0.25).

Genealogy — CLEAR framing

Predictive coding → AMI → I-JEPA → V-JEPA → 2-AC

Predictive coding

Rao & Ballard / Friston

EBM + AMI JEPA

LeCun [49][48]

Foils + tricks

MAE · data2vec · BYOL/DINO EMA

I-JEPA 2023

Image + multi-block

V-JEPA 2024

Video features

V-JEPA 2-AC

Robot CEM <62h DROID

Subsequent — CLEAR (official Meta follow-ons)

Cite these for video / robotics — not inventum of this PDF

V-JEPA

2404.08471 · Bardes et al. 2024

Same JEPA recipe on video; feature prediction; frozen probes K400 81.9 / SSv2 72.2

CLEAR

V-JEPA 2

2506.09985 · Assran et al. 2025

Internet-scale video JEPA + action-conditioned post-train; planning

CLEAR

V-JEPA 2-AC

same paper / facebookresearch/vjepa2

<62 h DROID → zero-shot Franka pick-place via CEM + image goals

CLEAR robotics

ijepa code

facebookresearch/ijepa

Official PyTorch ImageNet configs / checkpoints

CLEAR

TENTATIVE foils — design rhyme only

Do NOT cite as JEPA children / do not overclaim

DINOv2

TENTATIVE

Strongest modern discriminative dense features; complementary JEA foil — often paired with SigLIP in VLAs. Not JEPA lineage.

OpenVLA

TENTATIVE

Prismatic / SigLIP-DINOv2 fusion. No JEPA cite. Predictive pieces = AR action tokens, not latent JEPA.

BEHAVIOR winners

TENTATIVE

$\pi_{0.5}$ + FM + SigLIP stacks — parallel world-model bets, not V-JEPA 2-AC.

π_0 / $\pi_{0.5}$

TENTATIVE

SigLIP→PaliGemma + flow-matching action experts. Local packs: NO I-JEPA/V-JEPA cite. Architecture rhyme only.

GROOT-N1

TENTATIVE

Own dual-system; Eagle vision. Design rhyme with 'predict latent futures' — NOT JEPA lineage.

3D / other JEPA

TENTATIVE

Meta blog mentions — TENTATIVE pending pack-local PDFs.

Consistency triangle — use in every session

Three SSL families → three robotics bets

DINOv2 / SigLIP

Discriminative JEA
contrastive / sigmoid

→ VLA vision towers
(π_0 , OpenVLA, ...)

MAE

Generative MIM
pixels / tokens

→ strong FT
weaker linear probes

I-JEPA → V-JEPA → 2-AC

Predictive embedding
world-model line

→ planning robotics
(cite 2-AC)

Moral: VLAs chose SigLIP/DINO (left). JEPA's robotics payoff today is world-model / planning (right) — V-JEPA 2-AC — not replacing SigLIP in PaliGemma.

Predictor-as-world-model — Fig. 6 RCDCM

Pose / parts OK; discard unpredictable pixel detail — philosophical JEPA claim made visual

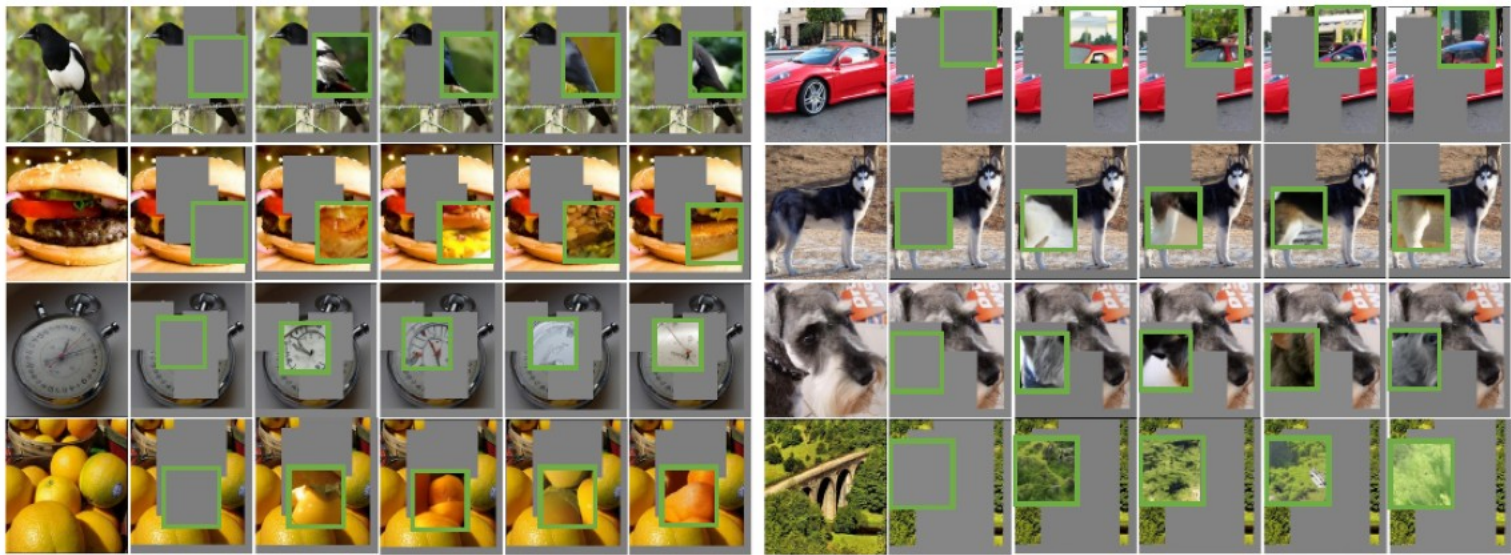


Figure 6. **Visualization of I-JEPA predictor representations.** For each image: first column contains the original image; second column contains the context image, which is processed by a pretrained I-JEPA ViT-H/14 encoder. Green bounding boxes in subsequent columns contain samples from a generative model decoding the output of the pretrained I-JEPA predictor, which is conditioned on positional mask tokens corresponding to the location of the green bounding box. Qualities that are common across samples represent information that contained is in the I-JEPA prediction. The I-JEPA predictor is correctly capturing positional uncertainty and producing high-level object parts with a correct pose (e.g., the back of the bird and top of a car). Qualities that vary across samples represent information that is not contained in the representation. In this case, the I-JEPA predictor discards the precise low-level details as well as background information.

Targets		Context		
Mask	Type	Erased	Type	Avg. Ratio* / Top-1
Paper Fig. 6 — qualitative only. Durable robotics idea: $P(\text{latent_target} \mid \text{latent_context}, z)$ without wasting capacity on pixel noise.				

Deepen toward robotics (Kanta)

I-JEPA idea $\neq$ robot numbers in this PDF

CLEAR claim

I-JEPA's durable robotics-facing idea is predictor-as-world-model: learn latent conditionals without pixel noise. V-JEPA 2-AC is the first CLEAR Meta demo of action-conditioned latent dynamics + zero-shot manip planning (<62 h DROID → Franka CEM / image goals).

For BEHAVIOR / VLA groups — open research

Ask whether an action-conditioned JEPA head on frozen video features beats / complements FM action experts. Not established in this 2023 paper. Cite 2-AC (2506.09985) for any robot success-rate claim — never this PDF's experiments.

Forbidden: 'I-JEPA achieves X% on Franka / BEHAVIOR.' Allowed: 'I-JEPA births the line that V-JEPA 2-AC later action-conditions.'

Must numbers — ablation + headline spine

Authoritative: ablations.md + Tables 1-7

54.2

T6 multi-block
vs ≤ 20.2 others

66.9 vs 40.7

T7 emb $\gg$ pixels
ViT-L/16 1% lin

72.4

T4 Clevr Dist
= MAE; $\gg$ DINO 53.4

79.3%

T1 H/14 linear
300 ep

73.3%

T2 H/14 1%
300 ep

81.1% / 77.3%

H/16@448
lin / 1%

$\sim 10\times$

vs MAE H/14
GPU-h

< 72 h / < 1200

16xA100 H/14
GPU-h

CLEAR vs TENTATIVE — label every claim

Exam discipline for lineage talks

CLEAR — say it

- LeCun AMI JEPA [48] → this paper
- Fig. 2/3/4 inventum mechanics
- V-JEPA 2404.08471 cites I-JEPA
- V-JEPA 2 / 2-AC 2506.09985
- Official ijepa / jepa / vjepa2 repos
- Ablation numbers T6-T12 from this PDF

TENTATIVE / FORBIDDEN

- π_0 /OpenVLA/GR00T 'are JEPA'
- I-JEPA robot SR / BEHAVIOR q-score
- DINOv2 is a JEPA child
- RCDM viz = quantitative proof
- 3D JEPA without local PDF
- Replacing SigLIP with I-JEPA in $\pi_{0.5}$ as established

Numbers tattoo — pin these

Shared across all five seats

79.3%

ViT-H/14 linear

73.3%

H/14 · 1% labels

81.1% / 77.3%

H/16@448 lin / 1%

54.2

multi-block (T6)

66.9 >> 40.7

emb >> pixels (T7)

72.4

Clevr Dist (T4)

~10×

cheaper than MAE H/14

<72 h

16×A100 H/14

<1200

GPU-hours H/14

87.1 vs 87.8

FT H/16@448 vs MAE

no view augs

single-view JEPA

2-AC not here

robots → cite 2506.09985

Archaeologist — close

Takeaways + exam bite

1. CLEAR inventum: image JEPA + multi-block + EMA + narrow predictor; no view augs.
2. CLEAR subsequent: V-JEPA → V-JEPA 2-AC — cite 2-AC for any robot SR.
3. Triangle: DINOv2/SigLIP | MAE | I-JEPA→2-AC — VLAs chose left; world-models chose right.
4. TENTATIVE: π_0 /OpenVLA/GR00T do not cite JEPA — design rhyme only.

EXAM BITE

I-JEPA = static-image birth certificate of Meta world-model line; actionable robotics = V-JEPA 2-AC (2506.09985), not this PDF.